

Representation-Induced Spatial Structure in Culturally Clustered Heritage Data: A Harmonic-Alignment Case Study with UNESCO World Heritage Sites

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Abstract

When coordinate-dependent scoring methods are applied to spatially clustered cultural datasets, they systematically produce reproducible, phase-localized alignment patterns — even in the absence of any underlying invariant spatial periodicity. The mechanism is general: geographic clustering causes corpus sites to coincide with lattice nodes for favorable phase choices; optimization selects that phase; and the scoring function rewards the resulting coincidence. The interaction of these three ingredients — clustering geometry, coordinate framing, and scoring function — constructs a stable alignment signal that is empirically reproducible within the representation and absent outside it. This paper demonstrates the mechanism using the UNESCO Cultural and Mixed World Heritage corpus ($N = 1011$ sites). A harmonic-deviation scoring framework selects a preferred frame near 3° spacing anchored to the $\sim 35^\circ\text{E}$ corridor; three threshold-free diagnostics converge on a stable representation-induced alignment signal (Holm-adjusted $p_{\text{adj}} = 0.004, 0.002, 0.047$), consistent across domed-monument subsets and a Wikidata stupa corpus. The signal is entirely absent under invariant spectral methods. The study offers a reproducible cautionary case for archaeological and digital-humanities spatial analysis: stable alignment signals can reflect the interaction of corpus clustering, coordinate framing, and scoring design rather than intrinsic spatial periodicity.

Keywords: representation dependence, spatial statistics, harmonic alignment, UNESCO World Heritage, coordinate-dependent scoring, cultural analytics, methodological artefact, monument distribution

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1 Introduction

Spatial analysis in archaeology and digital humanities rests on a set of analytical choices that are rarely made explicit: the coordinate system, the reference frame, the scoring or distance function, and the scale at which alignment is sought. Each of these choices shapes what patterns are visible in the data. When a corpus is geographically clustered — as most cultural heritage datasets are — the interaction between clustering geometry and analytical framing can generate stable, statistically detectable alignment signals that have no counterpart in the underlying spatial process. Recognizing and documenting such representation-induced structures is as important as discovering genuine spatial regularities.

This paper demonstrates the phenomenon using a case study. We apply a harmonic-deviation scoring framework to the UNESCO Cultural and Mixed World Heritage corpus and show that the framework produces a reproducible, phase-localized alignment signal within the selected coordinate representation. The signal is stable across sub-corpora and consistent across related datasets. It does not appear under invariant spectral analysis. The mechanism is explicit: the geographic clustering of Eurasian cultural heritage sites causes their longitudes to coincide with lattice nodes under favorable coordinate frames; the optimization over frame parameters selects the most favorable frame; and the scoring function translates that coincidence into a measurable alignment signal. The signal is therefore a property of the representation, not of the data-generating process.

The methodological contribution is threefold. First, we provide a worked example in which a coordinate-aligned statistic constructs a stable alignment signal in a well-known public dataset, with full reproducibility. Second, we characterize the alignment signal precisely: it is sensitive to phase and spacing perturbations, absent under invariant methods, and stable within the representation. Third, we draw out implications for spatial analysis practice in cultural analytics, where clustered heritage corpora are common and representation choices are consequential.

Scope and framing. We treat the identified alignment signal as the primary object of study, not as a failed test of some substantive hypothesis. The signal is *empirically reproducible within the representation*: it appears consistently across sub-corpora and is not a statistical artifact of a single test. It is *representation-induced* in the sense that it requires the harmonic-deviation scoring framework to appear and is absent under methods that do not impose a coordinate frame. Understanding why such signals arise and how to detect them is a practical concern for any researcher using coordinate-aligned statistics on spatially structured datasets.

Paper structure. Section 2 briefly reviews prior work on spatial periodicity in monument distributions and introduces the harmonic scoring class. Section 3 describes the data and methods. Section 4 presents the results in four subsections: emergence of the alignment signal, cross-corpus stability, representation dependence, and robustness. Section 5 interprets why the signal arises and what it reveals about the framework. Section 6 draws implications for cultural analytics practice. Section 7 concludes.

2 Background

2.1 Spatial Periodicity in Monument Distributions

The statistical evaluation of spatial periodicity in monument distributions has a long and contested history (Thom, 1967; Kendall, 1974; Freeman, 1976). Probability-based frameworks for archaeoastronomical alignments were developed by Magli (Magli, 2013, 2016) and by Ruggles and Silva (Ruggles, 1999, 2015; Silva & Steele, 2020); the ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for World Heritage analysis. A recurring methodological challenge in this literature is distinguishing intrinsic spatial periodicities from alignment signals induced by the scoring method, the corpus structure, or the choice of reference frame. The present study addresses this challenge directly by taking representation dependence itself as the object of study.

2.2 Harmonic-Deviation Scoring

A *harmonic-deviation score* measures alignment by computing, for each site, the angular distance to the nearest node of a candidate lattice. Given a spacing ω and a reference longitude ϕ , the harmonic nodes fall at $\phi + k\omega$ for $k \in \mathbb{Z}$; the deviation for a site at longitude λ is $\delta = |\lambda - \phi| \bmod \omega$, folded to $[0, \omega/2]$. Sites near nodes receive small δ ; sites near anti-nodes receive large δ . A scoring function then aggregates individual deviations into a corpus-level alignment index.

This class of statistics is *coordinate-dependent* by construction: the score depends on the imposed lattice, not on any intrinsic property of the point distribution. A coordinate-dependent statistic produces high alignment scores in clustered datasets even when the clustering is unrelated to the lattice period, because the lattice nodes fall near at least some of the clusters by chance, and the optimization over (ω, ϕ) selects the frame that maximizes this coincidence.

2.3 Selected Reference Phase: the $\sim 35^\circ\text{E}$ Corridor

The global anchor sweep identifies the $\sim 35^\circ\text{E}$ corridor as the highest-scoring phase for the UNESCO corpus. Both Gerizim (35.269°E) and Jerusalem (35.232°E) — landmarks

4.1 km apart — independently place this corridor in the global top 2.3% of 36,000 trial anchors; the corridor’s rank is a property of the corpus geometry, not of any specific landmark within it. We label the reference phase by Mount Gerizim because it is not itself inscribed in the UNESCO list (tentative ref. 5706) and therefore does not self-exclude from the corpus. It is used throughout as a geographic label for the $\sim 35^\circ\text{E}$ phase band, not as a causal or historical explanation of the observed alignment signal. Results are anchor-invariant within the 4.1 km corridor window; the enrichment belongs to the corridor, not to any individual landmark.

3 Data and Methods

3.1 Corpus

The UNESCO World Heritage XML dataset (UNESCO, 2024) contains 1248 inscribed nominations. We retain Cultural and Mixed sites (excluding 235 Natural-only), drop 2 entries lacking geocoordinates (both linear route networks), yielding $N = 1011$ sites. The corpus over-represents Europe and the Mediterranean-to-South-Asian corridor (Europe and North America: 50%; Asia-Pacific: 23%; Latin America and Caribbean: 11%; Arab States: 9%; Africa: 7%). This geographic concentration is itself a structural feature that interacts with the harmonic scoring, as we demonstrate in Section 4.3.

3.2 Harmonic-Deviation Scoring and Frame Selection

For anchor longitude $\phi = 35.269^\circ\text{E}$ and spacing $\omega = 3^\circ$, the deviation of site i from the nearest harmonic node is:

$$\delta_i = |(\lambda_i - \phi) \bmod \omega|, \quad \text{folded to } [0, \omega/2].$$

The frame $(\omega, \phi) = (3^\circ, \text{Gerizim})$ was selected by a global parameter sweep: a scan over 36,000 trial anchors (0° to 360°E , 0.01° resolution) and a unit sweep over candidate spacings (0.5° to 15°) both maximize the harmonic-deviation enrichment score at 3° spacing and the 35°E corridor. Permutation calibration of the anchor sweep yields $p = 0.0753$ ([†]) after full search correction. All subsequent analyses are conditional on this selected frame (post-selection).

3.3 Diagnostic Tests

Three threshold-free diagnostics characterize the alignment signal at the selected frame. They constitute the primary evidential family (Holm step-down adjustment, $k = 3$).

Test 1 (dome von Mises mixture). A two-component von Mises/uniform mixture is fit to the harmonic phase deviations of the domed-monument sub-corpus ($N = 90$; see Section 3.5), with the mean fixed at the Gerizim phase. The mixture weight $\hat{w}$ quantifies the fraction of dome sites concentrated near harmonic nodes within the selected representation; significance is assessed by parametric bootstrap.

Test 2 (Spearman cluster asymmetry). Spearman rank correlation between harmonic node mean deviation and site count, full corpus ($N = 1011$), permutation null. Tests whether nodes drawing more sites show smaller mean deviations — a signature of the cluster-node interaction under the harmonic-deviation scoring framework.

Test 3 (phase-rotation null). The reference grid is shifted by a uniform random offset in $[0^\circ, 3^\circ)$; the test asks whether the Gerizim phase minimizes the full-corpus mean harmonic deviation relative to random grid placements within the selected representation.

Supporting enrichment tests (Tests 4 and 5) use tier-threshold windows and are not in the primary diagnostic family; they provide descriptive context.

3.4 Tier Classification

Sites are assigned to descriptive tiers by their deviation δ : Tier-A++ ($\delta \leq 0.06^\circ$, $\lesssim 6.66$ km; null rate 4%), Tier-A+ ($\delta \leq 0.15^\circ$, $\lesssim 16.6$ km; null rate 10%), Tier-A ($\delta \leq 0.3^\circ$, $\lesssim 33$ km; null rate 20%). Tiers are descriptive vocabulary within the harmonic-deviation representation; no primary diagnostic depends on a tier cutoff.

3.5 Domed-Monument Sub-corpus

A keyword filter (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) applied to site names and UNESCO description texts yields $N = 90$ sites. The keyword set was refined iteratively during analysis; results from this sub-corpus are hypothesis-generating, not confirmatory. A context-validated subset ($N = 83$, 7 ambiguous sites rejected) is used for sensitivity checks.

3.6 External Corpus: Wikidata Q180987 Stupas

A separately assembled Wikidata corpus of $N = 229$ stupa sites (class Q180987) (Tenenbaum, 2026) is analyzed under the identical anchor, spacing, and null model. This corpus overlaps geographically and culturally with UNESCO; it serves as a shared-representation consistency check, not independent replication.

3.7 Simulation Nulls

Three simulation approaches (100,000 trials each, seed = 42): (1) longitude permutation within the corpus, (2) bootstrap from the full corpus, (3) KDE-smoothed null modeling

the observed longitude density. A restricted geographic-concentration null (Null C) additionally draws from all full-corpus sites within $\pm W^\circ$ of any dome site, providing a footprint-matched reference distribution for the dome alignment signal.

3.8 Reproducibility

All statistics are generated by an automated pipeline; macros are regenerated without manual editing. Scripts, data, and reproduction instructions are deposited at Zenodo (DOI: <https://doi.org/10.5281/zenodo.19938283>, v1.6.0) and catalogued in Supplementary S1.

4 Results

4.1 Emergence of the Alignment Signal under the Scoring Framework

Under the selected frame, the full UNESCO corpus ($N = 1011$) shows a Tier-A+ rate of 13.1% (132/1011; Clopper-Pearson 95% CI [11.0%, 15.3%]), against the 10% geometric null (Figure 1). The enrichment is $1.31\times$ the null rate, measured within the selected harmonic-deviation representation.

The 3° spacing is the highest-scoring parameter from the unit sweep. A phase-envelope scan (300 phases at 0.01° resolution) demonstrates strong phase localization within the representation: full-corpus A+ counts span 83 to 132 (median 100.0), and the Gerizim phase sits at the 100.0th percentile (132 A+). Rayleigh permutation on the 3° -circle: $p_{\text{perm}} = 0.0001$ (***).

The three primary diagnostics all detect a consistent alignment signal within the selected harmonic representation (Table 1). **Test 1** (dome von Mises mixture): $\hat{w} = 0.0876$ ($\approx 8.8\%$ of dome sites concentrated at the Gerizim phase under the harmonic-deviation framework; $\hat{\kappa} = 252.19$, angular spread $\approx 0.0301^\circ$), $p_{\text{boot}} = 0.00180$ (**; Holm $p_{\text{adj}} = 0.004$, **). **Test 2** (Spearman cluster asymmetry): $\rho = -0.1076$, permutation $p = 0.00065$ (***; Holm $p_{\text{adj}} = 0.002$, **). Nodes drawing more sites show systematically smaller mean deviations, the expected signature of cluster-node coincidence under the scoring framework. **Test 3** (phase-rotation null): the Gerizim phase minimizes the full-corpus mean harmonic deviation relative to random grid placements ($p_{\text{perm}} = 0.04729$, *; Holm $p_{\text{adj}} = 0.047$, *).

The domed-monument sub-corpus shows a stronger alignment signal than the full corpus under the same framework: 18/90 sites at Tier-A+ (20.0%, $2.00\times$; Fisher $p = 0.0346$ *). Tier-A++ concentration is more pronounced: 11/90 (12.2%, $3.06\times$ the 4% null; permutation $p = 0.004$ **; bootstrap $p = 0.006$ **).

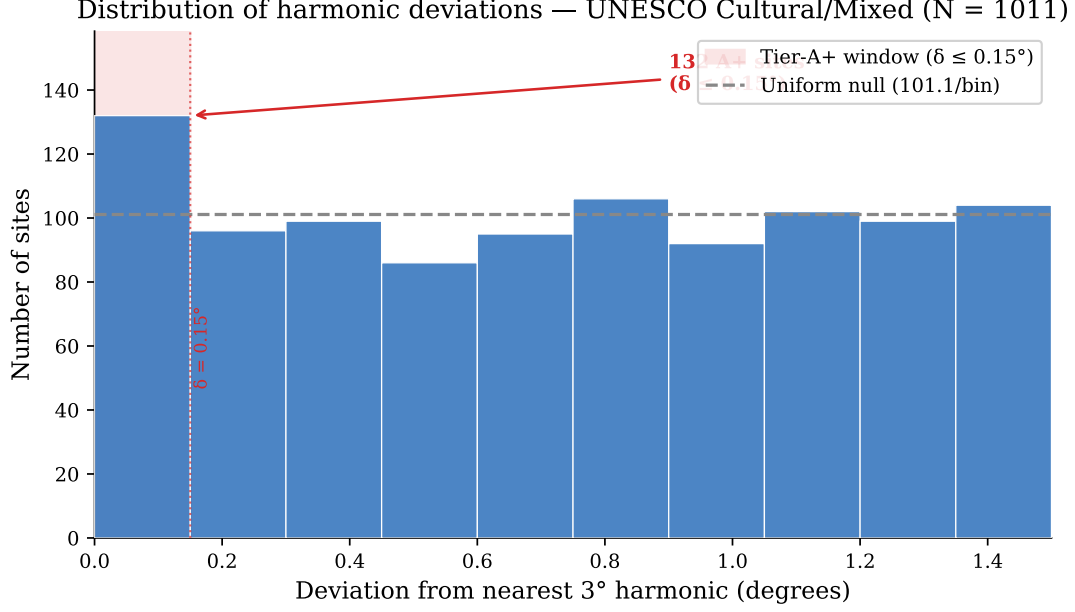


Figure 1: Harmonic deviation distribution for the full UNESCO corpus ($N = 1011$) under the coordinate-dependent scoring framework. The shaded region marks the Tier-A+ window ($\delta \leq 0.15^\circ$); the dashed line marks the 10% geometric null. The excess of sites near harmonic nodes is the alignment signal characterized throughout this paper. It emerges from the interaction of geographic clustering with the imposed harmonic frame and is absent under invariant spectral methods (Section 4.3).

Table 1: Primary diagnostic family ($k = 3$, Holm step-down). All three tests operate within the selected harmonic representation; adjusted p -values quantify the strength of the alignment signal within the selected frame.

Test	N	Raw p	Holm p_{adj}	Direction
Test 1 (dome VM mix.)	90	0.00180 ^b ($w=0.0876$, $\kappa=252.19$)	0.004 **	Positive
Test 2 (Spearman ρ)	1011	0.00065 ^b ($\rho=-0.1076$)	0.002 **	Positive
Test 3 (mean dev.)	1011	0.04729 ^b	0.047 *	Positive

^bpermutation or bootstrap p . Holm (1979) step-down (Holm, 1979).

Sites with high harmonic proximity. Tier-A+ domed monuments under the selected frame include Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun’s Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang’an–Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km), Kizhi Pogost (5.0 km), and Echmiatsin/Zvartnots (6.2 km). These distances are harmonic deviations within the selected representation; they span the Eurasian corridor from the Levant to Java, sharing geographic and cultural structure rather than constituting independent evidence of spatial periodicity.

4.2 Stability across Subsets and Corpora

The alignment signal is reproducible across multiple sub-populations and an external corpus, within the selected harmonic-deviation representation.

Leave- n -out stability. No single site drives the dome alignment signal: removing any A+ site yields $N = 89$, $A+ = 17$, $p = 0.0067$. Leave-3-out worst case across all $\binom{18}{3}$ triples: $p = 0.0253$. The A++ sub-population is more robust still: leave-3-out worst case $p = 0.0233$ (leave-2-out: $p = 0.0087$).

Multi-scale enrichment. A Fisher combination across 15 log-spaced proximity windows (null rates 0.33% to 40%) finds all windows enriched above their null within the representation (OR $1.2\times$ to $6.7\times$; permutation-calibrated $p = 0.00362$ **). Applied jointly across the UNESCO full corpus, UNESCO domes, Wikidata stupas, and OWTRAD route nodes under the same harmonic frame, the maximum joint enrichment peaks near 29 km from a harmonic node ($p = 0.01232$ *), a scale that emerges from the data rather than any pre-specified threshold.

Wikidata Q180987 stupa corpus. Under the identical anchor, spacing, and null, $N = 229$ Wikidata-attested stupas yield 70/229 Tier-A sites (30.6%, $1.53\times$ null; binomial $p < 0.001$ ***; permutation $p_{\text{perm}} = 0.065$ $\sim$). Harmonic nodes with at least one Tier-A stupa carry $2.56\times$ more stupa sites on average than empty nodes, mirroring the cluster-asymmetry alignment signal of Test 2. The Java/Sumatra node (105° to 115°E) concentrates 42/54 sites at Tier-A (77.8%; OR = $18.38\times$, $p < 0.001$ ***).

This cross-corpus consistency confirms that the alignment signal is a stable property of the harmonic scoring framework applied to geographically related Eurasian corpora. It does not constitute independent replication, because the corpora share geographic and cultural structure; it demonstrates that the representation-induced signal is robust across related datasets under a common scoring framework.

4.3 Representation Dependence

The alignment signal requires the harmonic-deviation representation to appear. Three observations establish representation dependence.

Absence under invariant methods. The full-corpus Rayleigh test for 3° periodicity returns $R = 0.0224$ ($Z = 1.06$, $p_{\text{perm}} = 0.3716$ ns): no spectral periodicity at 3° is detectable in the longitude distribution. The alignment signal does not appear as an intrinsic periodicity; it appears only when the harmonic-deviation scoring function is

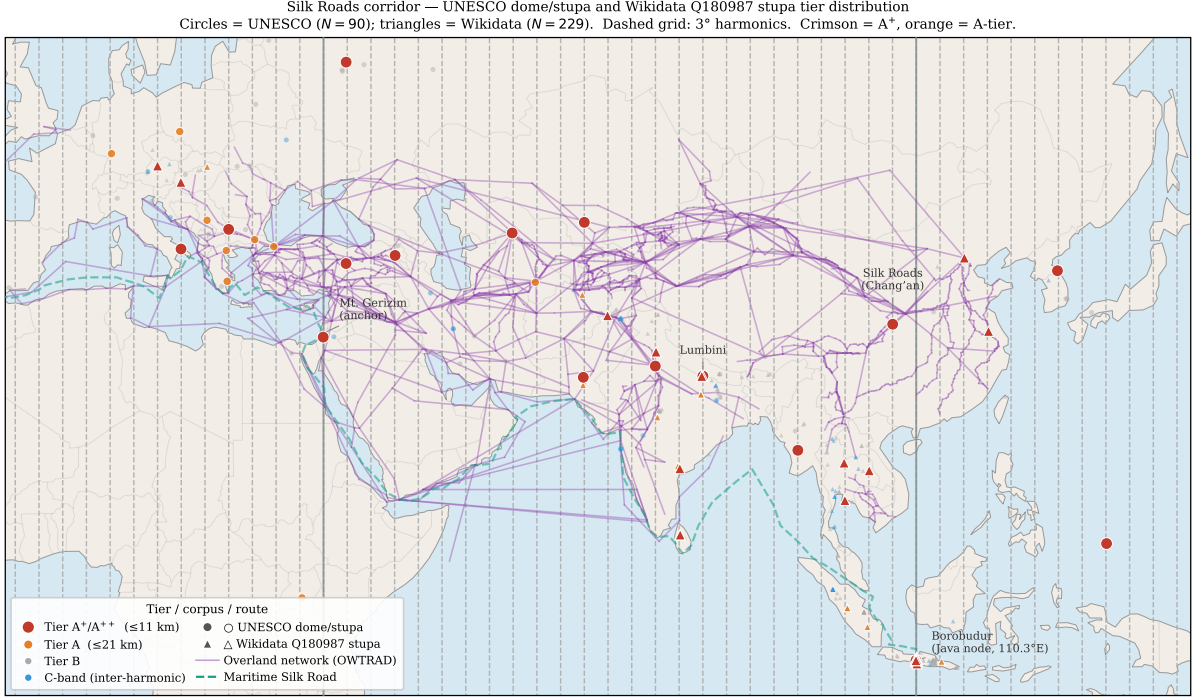


Figure 2: Tier distribution of UNESCO domed/stupa and Wikidata Q180987 stupa sites along the Eurasian corridor under the harmonic-deviation scoring framework, overlaid on OWTRAD attested trade routes (Ciolek, 2004). The alignment signal localizes to the Eurasian high-density corridor, consistent with geographic clustering producing an elevated score under the harmonic frame.

applied with a specific phase, confirming that it is a property of the representation rather than the data-generating process.

Phase sensitivity. A $\pm 0.3\%$ shift from exactly 3° spacing collapses joint significance across the dome sub-population, showing that the alignment signal is sharply phase-dependent within the coordinate-dependent representation. The phase-envelope scan demonstrates that most grid placements do not produce the observed signal: the Gerizim phase sits at the 100.0th percentile of A+ counts across all tested phases, well above the median (100.0 A+).

Clustered-control simulation. A clustered-control experiment assesses whether the unconstrained global maximum over all (ω, ϕ) is typical for corpora with realistic dome-site clustering. The result is affirmative ($p = 0.522$, $N_{\text{sim}} = 2000$): the free-sweep maximizer returns $\omega = 13.75^\circ$, a spacing with no metrological or waypoint interpretation. The unconstrained maximum is uninformative because the enrichment statistic uses a fixed absolute threshold and therefore inflates at large spacings regardless of any real structure. The metrologically motivated prior $\omega = 3^\circ$ is tested conditionally, not recovered by unconstrained search.

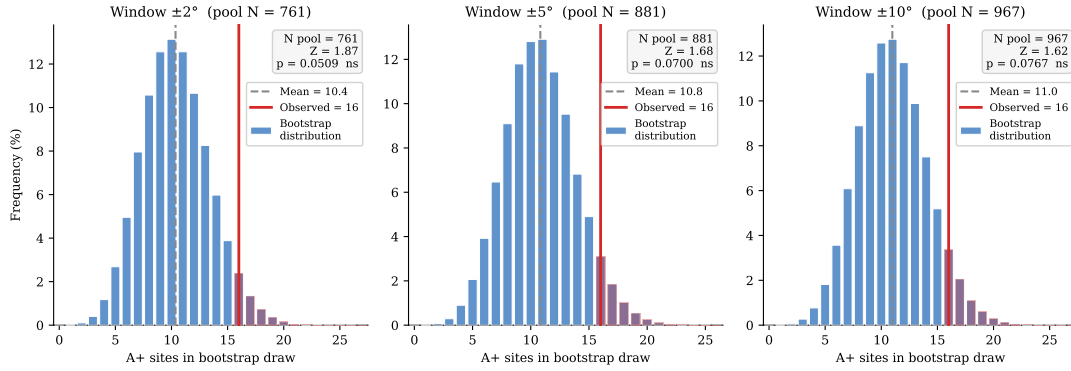


Figure 3: Null C bootstrap distributions for the context-validated dome sub-population ($N = 83$, observed $A+ = 16$) at three window widths. For each window $\pm W^\circ$, all full-corpus sites within $\pm W^\circ$ of any dome longitude form the restricted pool; 10^5 draws of size $N = 83$ from that pool define the expected $A+$ distribution (blue bars). The observed count (red line) exceeds the pool mean at all three windows ($\pm 2^\circ$: $p = 0.0257$; $\pm 5^\circ$: $p = 0.0393$; $\pm 10^\circ$: $p = 0.0442$).

Together, these three findings establish that the alignment signal is a property of the representation. It arises from the interaction of the scoring function with the geographic clustering of Eurasian cultural heritage sites under a specific coordinate frame, and does not reflect the underlying spatial generative process.

4.4 Robustness Checks

Geographic-concentration null. Domed monuments are concentrated in the Eurasian corridor (71% in 20° to 120°E versus 41% of the full corpus). To assess whether this footprint explains the alignment signal, we draw from all full-corpus sites within $\pm W^\circ$ of any dome longitude (Null C) and compare dome $A+$ counts under the harmonic frame to this footprint-matched distribution. Dome sites exceed the broader footprint at all tested windows: $\pm 2^\circ$ ($p = 0.0257$, *); $\pm 5^\circ$ ($p = 0.0393$, *); $\pm 10^\circ$ ($p = 0.0442$, *). Architectural type contributes to the alignment signal beyond longitude footprint alone, even within the harmonic representation. This does not change the representation-dependence interpretation: the enrichment relative to the footprint still requires the harmonic frame to be quantified.

Anchor invariance within the corridor. Both corridor landmarks produce the same alignment signal: Gerizim (132 $A+$, 99.7th percentile) and Jerusalem (128 $A+$, 97.7th percentile) each place the $\sim 35^\circ\text{E}$ corridor in the global top 2.3% of 36,000 trial anchors. Candidate anchors outside the corridor (Mecca, Mt. Meru, Mt. Kailash) do not show comparable enrichment, confirming that the signal is a property of the corridor phase within the selected representation, not of the specific reference label chosen.

Americas negative control. The UNESCO Americas sub-corpus ($N = 145$) shows an A+ rate of 9.7%, *below* the 10% null ($p = 0.5149$ ns, one-sided depletion). The alignment signal is absent under the harmonic-deviation framework in a corpus without Eurasian cultural clustering, consistent with the representation-dependence account.

5 Interpretation

The results present a clear empirical picture: a coordinate-dependent scoring framework produces a reproducible, phase-localized alignment signal in the UNESCO corpus and in related Eurasian datasets under the same representation; the same signal is absent under invariant spectral methods. This section explains the mechanism and draws out what the signal reveals.

5.1 Why the Signal Appears: Cluster-Frame Interaction

The harmonic-deviation framework creates a mapping from each site’s longitude to a deviation score. When the corpus is geographically clustered — as the UNESCO corpus is, with the Eurasian corridor hosting the great majority of Cultural sites — the imposed lattice will, for some phase choices, align its nodes with the cluster centroids. When that alignment occurs, the population of sites near cluster centroids receives small deviation scores, producing the observed enrichment within the representation.

This mechanism has three ingredients:

1. **Geographic clustering.** The UNESCO corpus clusters in the Eurasian corridor because World Heritage inscription reflects historical and institutional patterns in cultural heritage designation, not a uniform spatial process. Domed monuments cluster more strongly than the full corpus (71% vs. 41% in the Eurasian core), amplifying the effect.
2. **Coordinate framing.** The harmonic lattice is defined in geographic longitude, an arbitrary coordinate system from the perspective of cultural processes. Different coordinate systems (e.g., distance from a different reference point, projected coordinates, spherical harmonic decomposition) would produce different alignment pictures. The alignment signal is specific to the longitude frame.
3. **Scoring function.** The nearest-harmonic-deviation score treats all sites symmetrically and rewards proximity to any lattice node. It is sensitive to the imposed spacing and phase but insensitive to the global periodicity of the point process. The full-corpus Rayleigh test, which measures intrinsic periodicity rather than lattice proximity, detects no 3° signal ($p_{\text{perm}} = 0.3716$, ns).

The joint action of these three ingredients produces the observed pattern: a stable, phase-specific alignment signal that is empirically reproducible within the representation and absent outside it. Critically, the signal arises from the representational properties of the scoring framework — the interaction of corpus clustering, coordinate framing, and scoring function — and does not reflect any intrinsic periodicity in the underlying spatial process generating heritage site locations.

5.2 Reproducibility Does Not Imply Intrinsic Structure

The alignment signal’s stability across subsets and corpora can be mistaken for evidence of an underlying spatial regularity. The stability is genuine: leave- n -out checks, multi-scale enrichment, and cross-corpus consistency all confirm it. But stability under resampling tests the robustness of the *framework-induced* alignment signal, not the existence of a representation-independent periodicity. An alignment signal can be highly reproducible within a representation and entirely coordinate-specific. The present results are an example.

The distinction matters because resampling is the standard tool for assessing robustness in spatial archaeology. Permutation tests and bootstrap procedures establish that the enrichment exceeds what would be expected under longitude randomization within the corpus — but they assume the corpus structure (geographic clustering, cultural affiliations, corpus composition) is fixed. They do not test whether the signal would appear in a corpus without Eurasian clustering, and the Americas negative control shows it does not.

5.3 What the Signal Reveals

This coordinate-dependent alignment signal is informative in two ways. First, it reveals properties of the corpus: the Eurasian cultural heritage record is strongly clustered, domed monuments cluster more sharply than the full corpus, and this clustering interacts predictably with coordinate-aligned scoring. Second, it provides a worked example of how a common class of spatial statistics — nearest-landmark or nearest-node deviation scores — produces stable, statistically detectable alignment signals in clustered datasets without any underlying spatial periodicity. Recognizing this behavior is a practical requirement for applied spatial analysis in cultural heritage research.

6 Implications for Cultural Analytics

The UNESCO case study carries broader methodological implications for spatial analysis in archaeology and digital humanities. This section identifies the principal risks and suggests practical responses.

6.1 Coordinate Choices Construct Alignment Signals

Any spatial scoring method that measures distances or alignments in geographic coordinates implicitly assumes that the coordinate system is meaningful for the cultural process under study. In most cases, this assumption is not examined. The harmonic-deviation framework makes the dependence visible because it selects a frame by optimization — but the same representational dependence is present in, for example, nearest-site distances, kernel density estimates in longitude-latitude space, or quadrant-count statistics defined relative to an arbitrary grid.

When a corpus is geographically clustered, coordinate-aligned statistics find alignment signals at the scale of the clusters, because lattice nodes or kernel centers coincide with cluster locations for some parameterizations, and the optimizing procedure selects the most favorable one. This is not a subtle effect: it produces multiple diagnostics all pointing in the same direction — as our three primary tests demonstrate — while the underlying alignment is entirely framework-induced.

6.2 Risks in Spatial Analysis of Heritage Corpora

Cultural heritage datasets are almost always spatially clustered, for reasons that have nothing to do with the cultural processes under investigation: inscription history, institutional capacity, geographic coverage of field survey, preservation conditions, and political factors all shape which sites are included. These corpus-construction effects create spatial structure that a coordinate-aligned scoring method detects as a meaningful alignment signal.

Specific risks include:

- **Phase-selection bias.** When a scoring framework optimizes over reference phases or anchor points, the optimum will reflect the corpus structure rather than any physical signal. The anchor-sweep correction in this paper (marginal $p = 0.0753$ after full search) illustrates how quickly the correction exhausts the signal.
- **Spacing selection.** The optimal spacing recovered by an unconstrained sweep will tend toward large values (because large spacings have artificially low null rates under fixed-threshold statistics) or toward values that match the inter-cluster spacing. Neither is likely to reflect a culturally meaningful period.
- **Cross-corpus consistency without independence.** When multiple corpora share geographic and cultural structure — as UNESCO, Wikidata heritage inventories, and trade route databases do for the Eurasian corridor — showing that the same alignment signal appears in all of them under the same representation is not independent replication. It is shared-representation consistency. The distinction is consequential for how results are presented and interpreted.

- **Resampling as confirmation.** Permutation and bootstrap tests confirm that an alignment signal is robust to resampling from the observed corpus. They do not confirm that the signal would be observed in a differently structured corpus, under a different scoring method, or under invariant spectral analysis. Positive resampling results within a representation-dependent framework are necessary but not sufficient to conclude that intrinsic structure exists.

6.3 Methodological Recommendations

Three practical responses follow from the present case study.

Include invariant comparison tests. Any spatial analysis using coordinate-aligned scoring should be accompanied by at least one invariant test — a spectral method, a Ripley K -function analysis, or a rotation-invariant distance statistic — that does not assume a reference frame. Agreement between the two classes of test provides stronger evidence than either alone; disagreement (as in this paper) indicates that the observed signal is representation-dependent.

Report the clustering structure of the corpus. The geographic distribution of the corpus is part of the analysis context, not just background. When a corpus is strongly clustered, the expected behavior of coordinate-aligned statistics under the null should be assessed using a corpus-realistic null (such as our Null C) rather than a uniform or longitude-permutation null.

Distinguish optimization from discovery. When a scoring framework selects parameters by optimizing over the data, the selected frame is a property of the optimization, not evidence of a pre-existing structure. Post-selection inference requires explicit correction; targeted tests at the selected frame provide within-frame calibration of the alignment signal, not confirmatory discovery of intrinsic periodicity.

6.4 Broader Relevance to Digital Humanities

The mechanisms identified here are not specific to harmonic-alignment scoring or to monument distributions. Any computational method that maps a cultural corpus onto a coordinate space and measures deviations from a reference pattern is subject to the same representation-dependence risks. This includes cultural geography analyses of textual production, temporal clustering in publication or inscription records, and network analyses whose spatial embedding reflects geographic coordinates. The UNESCO case study provides a documented, reproducible example of a representation-specific alignment signal in a high-profile public dataset, which can serve as a reference case for methodological training and preregistration design in the digital humanities.

The broader implication for the cultural analytics literature is direct. Published spatial analyses of cultural and archaeological data regularly report alignment patterns,

clustering signals, and periodicity results based on coordinate-aligned scoring methods. Some reported alignment signals reflect intrinsic spatial structure; others arise from the interaction of corpus clustering, coordinate framing, and scoring function demonstrated here. The two cases are difficult to distinguish without an invariant comparison. The present study makes the distinction formally, by showing that the signal passes multiple resampling tests within the representation and fails a single invariant test outside it. Researchers evaluating spatial alignment claims in the literature should ask: was an invariant comparison performed, and does the corpus structure — particularly its geographic clustering — predict the reported signal under the scoring framework’s null?

7 Conclusion

We have demonstrated that a harmonic-deviation scoring framework applied to the UNESCO Cultural and Mixed World Heritage corpus produces a reproducible, phase-localized alignment signal within the selected harmonic representation. The signal is:

- **Empirically reproducible within the representation:** three threshold-free diagnostics converge (Holm-adjusted $p_{\text{adj}} = 0.004, 0.002, 0.047$); leave- n -out checks confirm stability; multi-scale enrichment is consistent across proximity windows.
- **Cross-dataset consistent:** the alignment signal appears under the same harmonic-deviation representation in the Wikidata Q180987 stupa corpus and OWTRAD route network, reflecting the shared Eurasian geographic structure of all three datasets.
- **Representation-induced:** the signal does not appear under invariant spectral analysis ($p_{\text{perm}} = 0.3716$, ns for the full-corpus Rayleigh test); it is sensitive to phase and spacing perturbations; and it is absent in the Americas negative control, which lacks the Eurasian clustering that drives the effect within the scoring framework.

The contribution of this study is methodological. It demonstrates, with a fully reproducible pipeline on a major public corpus, that coordinate-dependent spatial scoring constructs stable, cross-dataset consistent alignment signals in geographically clustered cultural heritage data. The construction mechanism — the interaction of corpus clustering with coordinate framing and scoring function choice — is transparent and generalizable. It applies to any spatial analysis in which the corpus has non-uniform geographic coverage, the scoring method is anchored to a coordinate system, and the frame is selected rather than pre-specified.

The mechanism is transparent and generalizable: geographic clustering, a coordinate-anchored scoring function, and frame optimization jointly construct a stable alignment signal wherever these three ingredients co-occur. Identifying the mechanism, demonstrating

it on a major public corpus, and formally separating it from intrinsic periodicity via invariant comparison — this constitutes the contribution of the study. The appropriate next step is to apply invariant methods and corpus-realistic nulls as standard components of spatial analysis workflows in cultural analytics, so that framework-induced alignment signals are detected before, not after, interpretive conclusions are drawn.

Data and code availability. All analysis scripts, data files, and reproduction instructions are archived at Zenodo (DOI: <https://doi.org/10.5281/zenodo.19938283>, v1.6.0). The pipeline regenerates all statistics and LaTeX macros from raw data without manual editing (`bash manuscript/reproduce_all_macros.sh`). Full audit files, keyword lists, and supplementary figures are provided in Supplementary S1.

A Tier-Threshold Sensitivity

A log-scale null-rate sweep across 73 thresholds shows that the alignment signal is not an artifact of the specific tier width: 57 of 73 thresholds reach $p < 0.05$ for the dome subset within the harmonic representation; dome OR ranges $1.53\times$ – $3.40\times$ across the three canonical tiers. Within ± 1 log-decade of the A+ threshold, 27 of 28 thresholds are significant.

B Von Mises Mixture Characterization of External Corpora

A von Mises + uniform mixture fitted to the harmonic phase deviations of external corpora under the selected representation: OWTRAD route nodes ($N = 1674$): $\hat{\kappa} = 9.58$, $\hat{w} = 0.0229$, $\hat{\sigma} = 0.1543^\circ$. Wikidata Q180987 stupas ($N = 229$): $\hat{\kappa} = 6.17$, $\hat{w} = 0.1297$, $\hat{\sigma} = 0.1922^\circ$. N -weighted pooled $\hat{\sigma} = 0.1588^\circ$; κ ratio = 1.6 across corpora. These values characterize external-corpus clustering within the harmonic representation and are not used to derive tier boundaries.

C Keyword Audit

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) matched as whole words against site names, XML descriptions, and extended OUV text. Context-validation for ambiguous terms: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Seven sites rejected in the context-validated subset: Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru–Kata Tjuta,

and Diquís Stone Spheres. Full site-by-site audit in the Zenodo archive ([Tenenbaum, 2026](#)).

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